

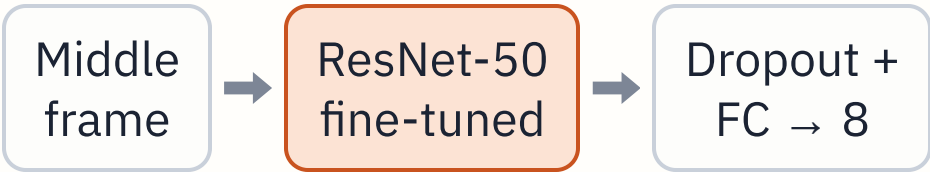
# From one frame to players and time

Test accuracy, 1,337 held-out clips  
Orange = new in this step · Blue = skip connection

## B1 · Single frame

62.6%

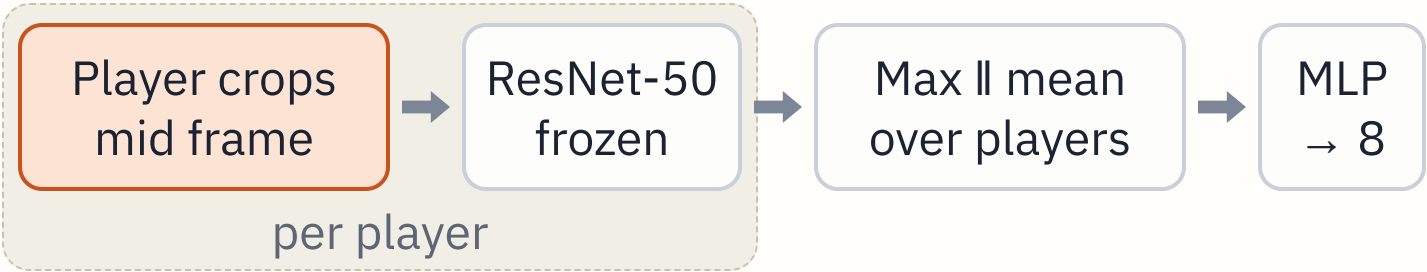
One middle frame, no players, no time. A fine-tuned ResNet-50: the floor every later model has to beat.



## B3 · Player crops

60.7%

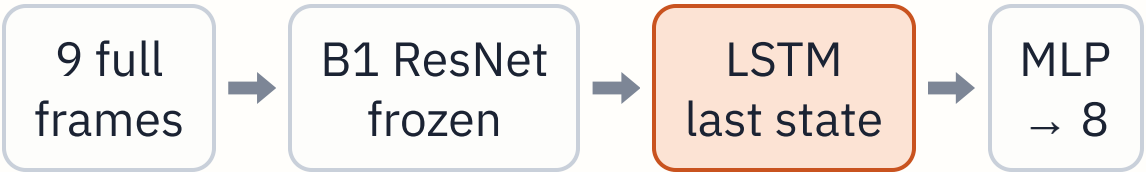
Adds players: a ResNet-50 fine-tuned on the 9 player actions, then frozen; its features are pooled across players. No time yet.



## B4 · Frames over time

66.1%

Adds time instead: B1's frozen ResNet on all 9 frames, an LSTM over them, and its last state classified. No players.



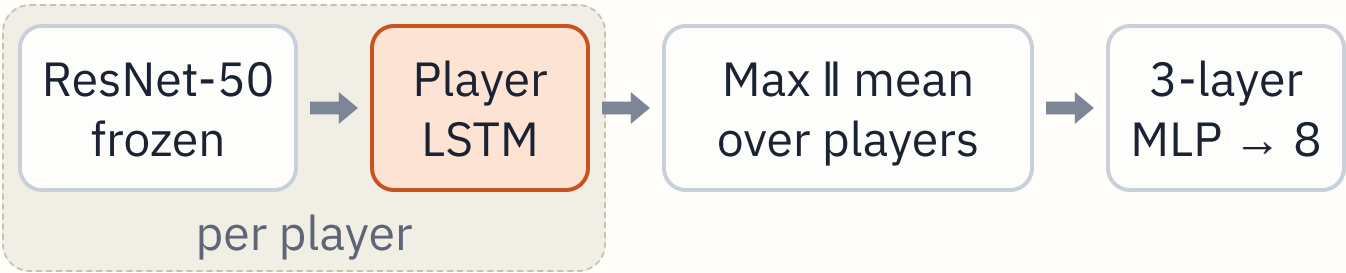
# Players over time, then the full hierarchy

Test accuracy, 1,337 held-out clips  
Orange = new in this step · Blue = skip connection

## B5 · Player LSTM

66.3%

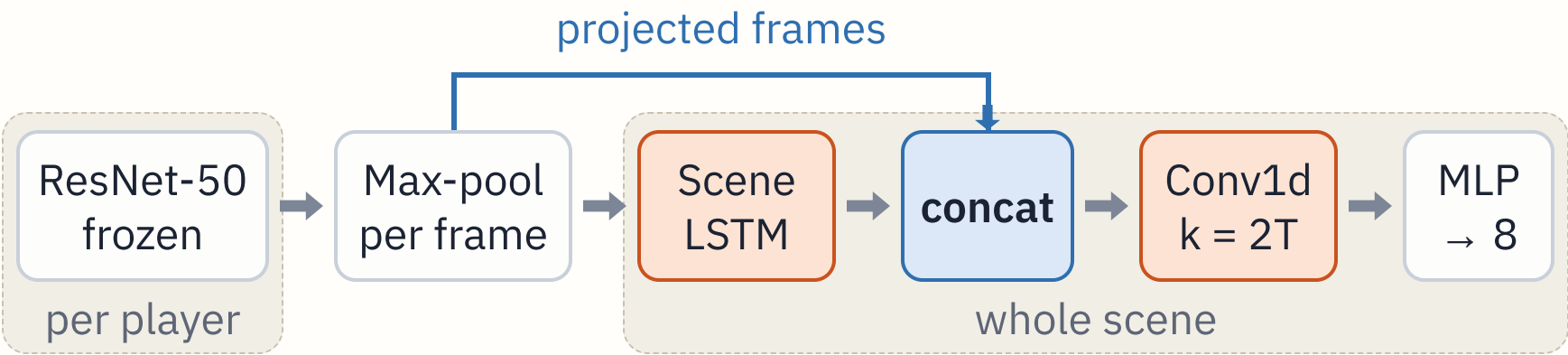
Each player's 9 frames go through a player LSTM pretrained on actions; its last states are pooled across players.



## B6 · Scene LSTM

70.5%

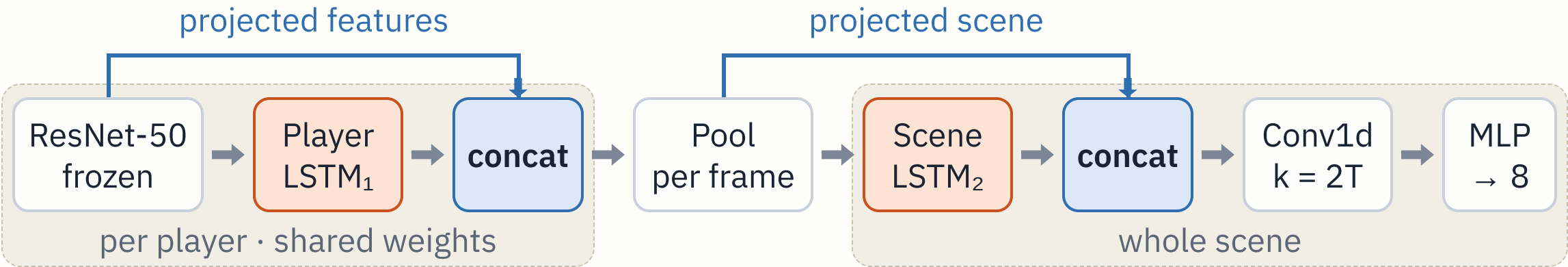
Pool players per frame, then a scene LSTM. A skip joins the projected frames to its 9 outputs; a Conv1d fuses all 18.



## B7 · Hierarchy

73.8%

The paper's two LSTMs plus two skips: player features join LSTM<sub>1</sub>'s output, and the pooled scene joins LSTM<sub>2</sub>'s.



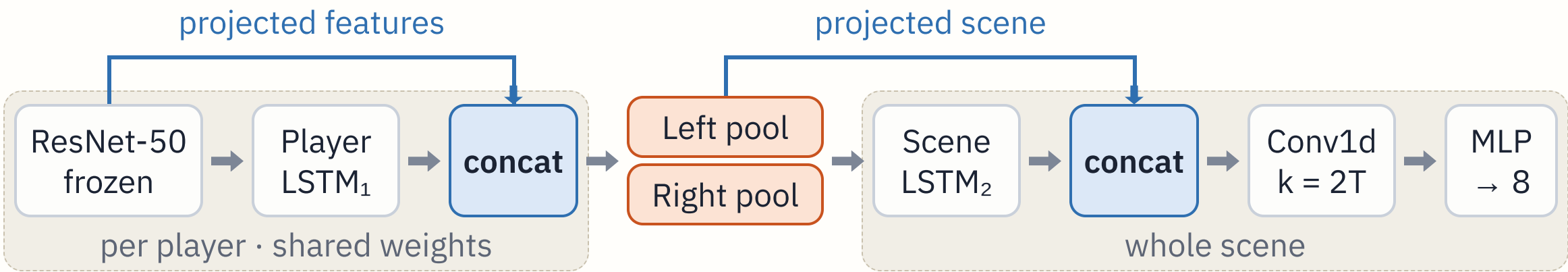
# One change wins: pool each team separately

Test accuracy, 1,337 held-out clips  
Orange = new in this step · Blue = skip connection

## B8 · Team split

85.6%

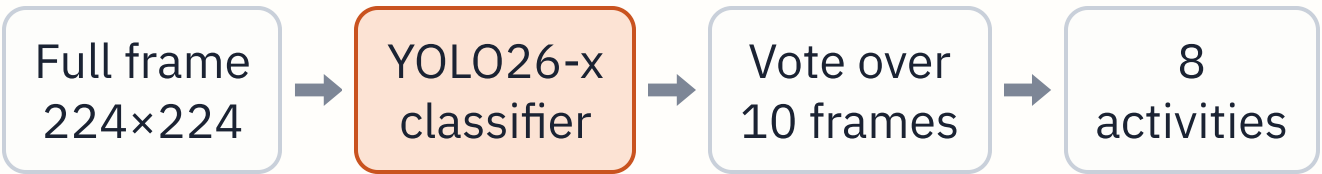
B7 with one change: each team is pooled on its own, so left and right stay apart. +11.9 points over B7.



## B9 · Control

80.3%

One YOLO classifier on the whole frame. No boxes, no tracks, no player labels; frames vote per clip.



85.6%

B8 beats the CVPR 2016 paper (81.9%) and stays about 5 points ahead of the frame-only B9, so player structure pays once the teams are kept apart.

Code: [github.com/omarTBakr/Volleyball-Group-Activity-Recognition](https://github.com/omarTBakr/Volleyball-Group-Activity-Recognition) · Demo: [huggingface.co/spaces/OmarTBakr/volleyball-activity-demo](https://huggingface.co/spaces/OmarTBakr/volleyball-activity-demo)